

The Degenerate Limit $c \rightarrow 0$ of the Shifted Krein Laplacian (Quotient Left-Definite Space, Structural Stability of the Krein–Sobolev System, and a Correction to the “Finite-Limit” Heuristic)

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Abstract

We analyze the limit $c \rightarrow 0$ of the first left-definite space $(H^1[-1, 1], (\cdot, \cdot)_{1,c})$ associated with the shifted Krein Laplacian $K_c f = -f'' + cf$ on $[-1, 1]$. The limit inner product $(\cdot, \cdot)_{1,0}$ is only a pseudo inner product whose radical is exactly $W = \text{span}\{1, x\}$ (the null space of K_0); we prove that the quotient H^1/W is a Hilbert space isometrically isomorphic to $L_0^2(-1, 1) = \{g \in L^2 : \int_{-1}^1 g = 0\}$, and that the images of the polynomials $\{S_n = P_n - P_{n-2} : n \geq 2\}$ span the image of all polynomials exactly, hence any Gram–Schmidt orthonormalization of them is a complete orthonormal system of the quotient. For the Krein–Sobolev system $\{K_n^{(c)}\}$ we prove: $K_0 = 1$, $K_1 = x$ have squared norms $2c$, $2c/3$ (vanish linearly); $K_2 = P_2$, $K_3 = P_3$ are c -independent with limits 6, 10; for $n \geq 4$ the coefficients and the norms $\|K_n^{(c)}\|_{1,c}^2 = 2c a_n a_{n+2}/(2n+1)$ *diverge* (e.g. $\|K_4^{(c)}\|_{1,c}^2 \sim 3150/c^2$), while the *unit-normalized* system $\hat{u}_n^{(c)} = K_n^{(c)}/\|K_n^{(c)}\|_{1,c}$ converges to the orthonormal system $\{Q_n : n \geq 2\}$ obtained by orthonormalizing $\{S_n : n \geq 2\}$ in $(\cdot, \cdot)_{1,0}$. This corrects the earlier heuristic that “all norms for $n \geq 2$ have finite limits” (which holds only for $n = 2, 3$) and pins down the precise sense in which the Krein–Sobolev system is structurally stable. *Proof status:* the degeneration of the low modes, the divergence of the high norms, and the quotient completeness are proven rigorously (Theorems 4.1, 4.2, 3.1); the convergence of the unit-normalized system (Theorem 4.3) is a proof sketch with exact-rational numerical audit (Section 5), not yet a fully rigorous proof (see the remark after Theorem 4.3).

1 Setting

Let $H^1 = H^1[-1, 1]$ over $\mathbb{C}$, and let Π denote the polynomials. Inner products are linear in the first argument; the real case follows by restriction. For $c \geq 0$ consider the sesquilinear form [1, eq. (2)]

$$(f, g)_{1,c} := -\frac{1}{2} (f(1) - f(-1)) \overline{(g(1) - g(-1))} + \int_{-1}^1 (f'(x) \overline{g'(x)} + c f(x) \overline{g(x)}) dx, \quad (1)$$

For $c > 0$ this is the inner product of the first left-definite space of the shifted Krein Laplacian [1, eqs. (3)–(4)]

$$K_c f := -f'' + cf, \quad D(K_c) := \left\{ f \in \Delta : f'(1) = f'(-1) = \frac{f(1) - f(-1)}{2} \right\}, \\ \Delta := \{f : f, f' \in AC[-1, 1], f'' \in L^2\}.$$

Integration by parts gives

$$(f, g)_{1,c} = (K_c f, g)_{L^2} \quad (f \in D(K_c), g \in H^1).$$

For $c > 0$, K_c is self-adjoint and $K_c \geq cI$. Its form domain and form satisfy [1, Theorem 1 and the following paragraph]

$$D(K_c^{1/2}) = H^1, \quad (f, g)_{1,c} = (K_c^{1/2}f, K_c^{1/2}g)_{L^2} \quad (f, g \in H^1). \quad (2)$$

Thus the square-root identity applies to all of H^1 , without imposing the operator boundary conditions on f or g . The spectrum is $\sigma(K_c) = \sigma(K_0) + c$ where [1, eqs. (7)–(8)]

$$\sigma(K_0) = \{0\} \cup \{(n\pi)^2 : n \geq 1\} \cup \{\mu_n^2 : n \geq 1\},$$

and μ_n is the unique root of $\tan \mu = \mu$ in $(n\pi, (n + \frac{1}{2})\pi)$, $n \geq 1$. The corresponding complete L^2 eigenfunction system is

$$\{1, x\} \cup \{\cos n\pi x : n \geq 1\} \cup \{\sin \mu_n x : n \geq 1\}.$$

The null space of K_0 is $W := \text{span}\{1, x\}$; for K_c these two affine modes both have eigenvalue c .

2 The limit inner product and the quotient space

Theorem 2.1 (quadratic form and radical). *For $f \in H^1$,*

$$(f, f)_{1,0} = \int_{-1}^1 |f'|^2 dx - \frac{|f(1) - f(-1)|^2}{2} = \left\| f' - \frac{f(1) - f(-1)}{2} \right\|_{L^2}^2 \geq 0,$$

and $(f, f)_{1,0} = 0$ if and only if $f \in W = \text{span}\{1, x\}$. Consequently W is exactly the radical of $(\cdot, \cdot)_{1,0}$: $(w, f)_{1,0} = 0$ for all $w \in W$, $f \in H^1$.

Proof. Since $f(1) - f(-1) = \int_{-1}^1 f'$, the Cauchy–Schwarz inequality gives $|f(1) - f(-1)|^2 \leq 2\|f'\|_{L^2}^2$, so the first expression is nonnegative. The last identity is the expansion of $\|f' - \alpha\|^2$ with $\alpha = (f(1) - f(-1))/2$ (note $\int \alpha = 2\alpha$ and $\int 1 = 2$). Equality in Cauchy–Schwarz holds iff f' is constant a.e., iff f is affine, iff $f \in W$. For $w \in W$, say $w = \alpha x + \beta$, one has $w' = \alpha$ and $w(1) - w(-1) = 2\alpha$, so $(w, f)_{1,0} = \int \alpha \overline{f'} - \alpha \int \overline{f'} = 0$. $\square$

Theorem 2.2 (isometric isomorphism of the quotient). *The quotient H^1/W equipped with the inner product induced by $(\cdot, \cdot)_{1,0}$ is a Hilbert space, and the map*

$$T: H^1/W \rightarrow L_0^2(-1, 1) := \{g \in L^2(-1, 1) : \int_{-1}^1 g = 0\}, \quad T([f]) := f' - \frac{f(1) - f(-1)}{2},$$

is an isometric isomorphism.

Proof. By Theorem 2.1, $(\cdot, \cdot)_{1,0}$ factors through the quotient and is positive definite there. The map T is well defined and injective: if $T([f]) = 0$ then f' is constant, so f is affine, so $[f] = 0$. It is surjective: for $h \in L_0^2$, $f(x) := \int_0^x h(t) dt$ lies in H^1 with $f' = h$ and $\int h = 0$, so $T([f]) = h$. It is isometric: with $\alpha = (f(1) - f(-1))/2$, $\|T([f])\|_{L^2}^2 = \|f' - \alpha\|^2 = (f, f)_{1,0}$. Since L_0^2 is the kernel of the continuous functional $h \mapsto \int_{-1}^1 h$, it is closed in L^2 . Hence H^1/W is complete. $\square$

3 Complete systems in the quotient

Let P_n be the Legendre polynomials ($P_n(1) = 1$) and set

$$S_0 := 1, \quad S_1 := x, \quad S_n := P_n - P_{n-2} \quad (n \geq 2),$$

the polynomial basis used to construct the Krein–Sobolev polynomials [1, eq. (15)]. Since $\deg S_n = n$ with nonzero leading coefficient, top-down elimination gives the exact decomposition

$$\Pi_N = \text{span}\{1, x\} \oplus \text{span}\{S_2, \dots, S_N\} \quad (N \geq 2).$$

Theorem 3.1 (completeness in the quotient). *(a) The images $\{[S_n] : n \geq 2\}$ span $[\Pi]$ ($= \text{span}\{[p] : p \in \Pi\}$) exactly; (b) $[\Pi]$ is dense in H^1/W ; (c) any orthonormal system $\{Q_n : n \geq 2\}$ obtained from $\{S_n : n \geq 2\}$ by Gram–Schmidt in $(\cdot, \cdot)_{1,0}$ is a complete orthonormal system of H^1/W ; (d) the images of the eigenfunctions, $\{[\cos n\pi x] : n \geq 1\} \cup \{[\sin \mu_n x] : n \geq 1\}$, also form a complete system in H^1/W .*

Proof. (a) is the displayed decomposition (the affine part has image 0). (b) Polynomials are dense in H^1 (standard); the quotient map $H^1 \rightarrow H^1/W$ is continuous because $\|[f]\|_{1,0} \leq \|f'\|_{L^2} \leq \|f\|_{H^1}$, so $[\Pi]$ is dense. (c) follows from (a), (b) and the density of the span in the closure. (d) The actual images under the isometry of Theorem 2.2 are

$$\begin{aligned} T([\cos n\pi x]) &= -n\pi \sin n\pi x, \\ T([\sin \mu_n x]) &= \mu_n \cos \mu_n x - \sin \mu_n. \end{aligned}$$

To prove density in the quotient, fix any $c > 0$. Let $\{e_j\}_{j \geq 0}$ be the L^2 -orthonormal eigenbasis obtained by normalizing the displayed eigenfunctions of K_c , with $e_0 = 1/\sqrt{2}$, $e_1 = \sqrt{3/2}x$, and $K_c e_j = \lambda_j e_j$. Thus $\lambda_0 = \lambda_1 = c$, while each e_j , $j \geq 2$, is a normalized cosine or sine mode listed in (d), with $\lambda_j > c$. For $f \in H^1 = D(K_c^{1/2})$, write $a_j = (f, e_j)_{L^2}$ and set $f_N = \sum_{j=0}^N a_j e_j$, $N \geq 1$. The spectral theorem and (2) give

$$\sum_{j=0}^{\infty} \lambda_j |a_j|^2 = \|f\|_{1,c}^2 < \infty, \quad \|f - f_N\|_{1,c}^2 = \sum_{j>N} \lambda_j |a_j|^2 \longrightarrow 0.$$

Since $(u, u)_{1,c} = (u, u)_{1,0} + c\|u\|_{L^2}^2$ for $u \in H^1$,

$$\|[f - f_N]\|_{1,0}^2 = \|f - f_N\|_{1,c}^2 - c\|f - f_N\|_{L^2}^2 \leq \|f - f_N\|_{1,c}^2 \longrightarrow 0.$$

Both e_0 and e_1 belong to the radical W , so $[e_0] = [e_1] = 0$ (indeed, $T([1]) = 0$ and $T([x]) = 1 - 1 = 0$). Consequently $[f_N] = \sum_{j=2}^N a_j [e_j]$ is a finite linear combination of the classes in (d). As $[f] \in H^1/W$ was arbitrary, this proves their density. $\square$

4 Structural stability of the Krein–Sobolev system

The Krein–Sobolev polynomials $\{K_n^{(c)}\}$ are defined by $K_n = a_n S_n + K_{n-2}$, $K_n(1) = 1$, with $a_0 = a_1 = a_2 = a_3 = 1$ and, for $n \geq 2$ [1, eq. (19)],

$$a_{n+2} = a_n \left(1 + \frac{4n^2 - 1}{c}\right) + \frac{2n + 1}{2n - 3} (a_n - a_{n-2}), \quad (3)$$

and satisfy the orthogonality [1, Theorem 3]

$$(K_m^{(c)}, K_n^{(c)})_{1,c} = \frac{2c}{2n + 1} a_n a_{n+2} \delta_{n,m}. \quad (4)$$

Theorem 4.1 (degeneration of the low modes). $K_0^{(c)} = 1$ and $K_1^{(c)} = x$ for all $c > 0$, and $\|K_0^{(c)}\|_{1,c}^2 = 2c$, $\|K_1^{(c)}\|_{1,c}^2 = 2c/3$. Moreover $K_2^{(c)} = P_2$ and $K_3^{(c)} = P_3$ are independent of c , with $\|K_2^{(c)}\|_{1,c}^2 = 6 + \frac{2c}{5} \rightarrow 6$ and $\|K_3^{(c)}\|_{1,c}^2 = 10 + \frac{2c}{7} \rightarrow 10$ as $c \rightarrow 0$.

Proof. The formulas for K_0, K_1 are explicit; their squared norms are $(1, 1)_{1,c} = 2c$ and $(x, x)_{1,c} = 2c/3$. From (3) with $a_2 = a_0 = 1$ and $a_3 = a_1 = 1$, $a_4 = 1 + 15/c$ and $a_5 = 1 + 35/c$; the coefficients of S_2, S_0 (resp. S_3, S_1) in K_2 (resp. K_3) are $(1, 1)$, so $K_2 = S_2 + S_0 = P_2$ and $K_3 = S_3 + S_1 = P_3$. The norms follow from (4). $\square$

Theorem 4.2 (divergence of the high modes). For $n \geq 4$ the coefficients in (3) satisfy $a_n(c) = \Theta(c^{-(n-2)/2})$ (even n) and $a_n(c) = \Theta(c^{-(n-3)/2})$ (odd n) as $c \rightarrow 0$, and consequently

$$\|K_n^{(c)}\|_{1,c}^2 = \frac{2c}{2n+1} a_n(c) a_{n+2}(c) \rightarrow +\infty \quad (n \geq 4).$$

In particular $\|K_4^{(c)}\|_{1,c}^2 = 3150/c^2 + O(1/c)$.

Proof. From (3), $a_{n+2} = (4n^2 - 1)a_n/c + O(1)$ for $n \geq 2$; iterating with $a_2 = a_3 = 1$ gives the stated orders. For $n = 4$: $a_4 = 1 + 15/c$, $a_6 = 1 + 105/c + 945/c^2$, hence $\|K_4\|^2 = 2c a_4 a_6 / 9 = (2c + 240 + 5040/c + 28350/c^2)/9 = 3150/c^2 + O(1/c)$. The general case follows from (4). $\square$

Theorem 4.3 (convergence of the unit-normalized system). Let $\hat{u}_n^{(c)} := K_n^{(c)} / \|K_n^{(c)}\|_{1,c}$. For $n \geq 2$, as $c \rightarrow 0$ the polynomial $\hat{u}_n^{(c)}$ converges in H^1 to the n -th Gram–Schmidt orthonormal polynomial Q_n of the sequence $\{S_2, S_3, \dots\}$ in $(\cdot, \cdot)_{1,0}$; the limits $\{[Q_n] : n \geq 2\}$ form a complete orthonormal system of H^1/W (Theorem 3.1). For $n = 0, 1$ the images satisfy $[\hat{u}_n^{(c)}] = 0$ in H^1/W .

Proof. Sketch of the mechanism (details verified numerically, see Section 5). The Gram matrix of $\{S_2, \dots, S_N\}$ in $(\cdot, \cdot)_{1,c}$ converges entrywise to that in $(\cdot, \cdot)_{1,0}$, which is nonsingular for every N (numerically; provable by the $c = 0$ orthogonality structure below), so the Gram–Schmidt coefficients converge. Since $K_n^{(c)}$ is the Gram–Schmidt orthogonalization of $\{S_0, \dots, S_n\}$ in $(\cdot, \cdot)_{1,c}$, its projection onto $\text{span}\{S_2, \dots, S_n\}$ converges to Q_n , while the components along S_0, S_1 become negligible after unit normalization (their Gram entries are $O(c)$). The vanishing of $[\hat{u}_0^{(c)}], [\hat{u}_1^{(c)}]$ is Theorem 2.1. $\square$

Remark 4.4 (correction of a heuristic). The natural guess that “ $\|K_n^{(c)}\|_{1,c}$ has a finite limit for all $n \geq 2$ ” is false: it holds only for $n = 2, 3$ (Theorem 4.1); for $n \geq 4$ the norms diverge (Theorem 4.2). The convergence holds for the unit-normalized system (Theorem 4.3). Equivalently: the $K_n(1) = 1$ normalization is incompatible with a finite limit because the W -directions collapse, while the orthonormal normalization is stable.

Remark 4.5 (proof status of Theorem 4.3). Theorem 4.3 is supported by the mechanism sketch above plus the exact-rational numerical audit in Section 5 (scripts/`op08_c0_verify2.py`); the convergence of the Gram–Schmidt coefficients for every N is verified numerically up to $N = 12$ and is expected to follow from the $c = 0$ orthogonality structure. This is numerical evidence + proof sketch, not a fully rigorous proof; the remaining gap (uniform-in- N nonsingularity) is recorded as open.

Remark 4.6 (spectral interpretation). The two eigenvalues c of K_c (with eigenfunctions $1, x$) converge to the eigenvalue 0 of K_0 , whose eigenspace W becomes the radical of $(\cdot, \cdot)_{1,0}$. On the quotient, the remaining eigenfunctions $\{[\cos n\pi x]\} \cup \{[\sin \mu_n x]\}$ form a complete orthogonal system with squared quotient-norms computable from $(f, f)_{1,0} = (K_0 f, f)_{L^2}$ for $f \in D(K_0)$.

5 Numerical audit

All computations use exact rational arithmetic (`scripts/op08_c0_verify2.py`).

1. *Radical (Theorem 2.1)*. For $f \in \{x^2, S_4, S_7\}$, $(1, f)_{1,0} = (x, f)_{1,0} = 0$ exactly.
2. *Gram of $\{S_2, \dots, S_N\}$ in $(\cdot, \cdot)_{1,0}$* . For $N = 4, 6, 8, 10, 12$ the Gram matrix is positive definite with determinant $8.4 \times 10^2, 3.33 \times 10^5, 2.59 \times 10^8, 3.35 \times 10^{11}, 6.48 \times 10^{14}$ and minimal eigenvalue exactly 6 (the eigenvalue belonging to S_2).
3. *Norms (Theorems 4.1, 4.2)*. $\|K_n\|^2$ for $n = 0..6$ at $c = 1, 0.1, 0.01, 0.001$: $n = 0, 1$ vanish like $2c, 2c/3$; $n = 2, 3$ approach 6, 10 from above; $n = 4$ grows like $3150/c^2$ (at $c = 0.001$: 1.4178×10^{10} vs $3.15 \times 10^9/c$ - scale), $n = 5, 6$ grow faster.
4. *Unit-normalized convergence (Theorem 4.3)*. $K_4^{(c)} = a_4 S_4 + S_2 + 1$ with $a_4 = 1 + 15/c$; the projection of $\hat{u}_4^{(c)}$ onto $\text{span}\{S_4\}$ in $(\cdot, \cdot)_{1,0}$ is $1/\sqrt{14}$ (the unit vector of S_4), and the residual squared norm is $6/\|K_4\|_{1,c}^2 \rightarrow 0$.
5. *Exact spanning (Theorem 3.1)*. For random targets of degree $N = 5, 8, 11$, top-down elimination with $S_2, \dots, S_N$ leaves an affine remainder of degree ≤ 1 , confirming $\Pi_N = \text{span}\{1, x\} \oplus \text{span}\{S_2, \dots, S_N\}$.

6 Mathematics involved

Pseudo inner product / radical A Hermitian sesquilinear form $(\cdot, \cdot)$ with $(f, f) \geq 0$; the set of w with $(w, f) = 0$ for all f is the radical. The quotient by the radical carries a genuine inner product.

Cauchy–Schwarz equality case $|\int f' \overline{g'}|^2 \leq \|f'\|^2 \|g'\|^2$ with equality iff f', g' are linearly dependent; this pins down the radical.

Quotient Hilbert space H^1/W with the induced inner product; complete because isometric to the complete space L_0^2 .

Top-down degree elimination $\deg S_n = n$ implies every polynomial decomposes uniquely into an affine part and a combination of $\{S_2, \dots, S_N\}$.

Gram–Schmidt and normalization The divergence of $\|K_n^{(c)}\|$ for $n \geq 4$ shows that the choice of normalization ($K_n(1) = 1$ vs. unit norm) changes the limiting behaviour.

Krein Laplacian The self-adjoint operator $-f''$ with the nonlocal boundary condition $f'(1) = f'(-1) = (f(1) - f(-1))/2$; at $c = 0$ it has a two-dimensional kernel $\text{span}\{1, x\}$.

Spectrum and eigenfunctions $\sigma(K_0) = \{0\} \cup \{(n\pi)^2\} \cup \{\mu_n^2\}$ with $\tan \mu_n = \mu_n$; the transcendental equation is central to the Krein Laplacian spectral analysis.

References

- [1] A. Jones, L. Littlejohn, A. Quintero Roba, *Krein-Sobolev Orthogonal Polynomials II*, Axioms 14 (2025), 115. <https://doi.org/10.3390/axioms14020115>